Practice Set 14 Paper 1 – Basic Sciences in Urology

1. Q1. Developmental embryology of the testes, physiological descent and abnormalities of descent (LAQ)
2. Q2. Staging of CA prostate and management of non-organ-confined CA prostate (LAQ)
3. Q3a. Testicular microlithiasis (SAQ)
4. Q3b. Postoperative lymphatic leak (SAQ)
5. Q3c. Hyperfiltration injury in AKI and CKD (SAQ)
6. Q3d. Pelvicalyceal anomalies other than PUJ obstruction (SAQ)
7. Q3e. TURP instruments (SAQ)
8. Q3f. Methods to estimate prostate size (SAQ)

Practice paper compiled from the question bank.

Q1. Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent.

Type: LAQ (25 marks) (descent also answered in 2024-Winter Paper 1, Q2; maldescent in Practice Set 5 Paper 3, Q3b)

Development of the testis

- **Indifferent gonad:** the gonadal (germinal) ridge forms at **4 weeks** on the medial urogenital ridge. It has four components: coelomic epithelium, underlying mesenchyme, adjacent mesonephros and **PGCs (primordial germ cells)**.
- PGCs arise in the **yolk sac wall in week 3** and migrate along the dorsal mesentery of the hindgut to the ridge (about 6 weeks).
- **SRY (sex-determining region Y)** on the short arm of Y, partly activated by SF1 (steroidogenic factor 1), switches on **SOX9** in pre-Sertoli cells. Sertoli cells then appear (about 40 days). Without SRY no Sertoli cells form and the gonad becomes an ovary.
- Sertoli cells enclose germ cells as **testicular cords (about 6 weeks)**, which canalise into seminiferous tubules.
- **Rete testis** forms with the cords and joins **5–12 mesonephric tubules**, which become efferent ductules. Full continuity (tubules, rete, efferent ductules, epididymis) is established by the **third month**.
- The testis turns round and a thick **tunica albuginea** forms. Testes become spherical after 7–8 weeks; ovaries stay oblong.

Hormone	Cell	Action	Male remnants
AMH (anti-Müllerian hormone)	Sertoli	Müllerian regression, weeks 8–10	Appendix testis, prostatic utricle
Testosterone	Leydig (from 8 weeks; driven by placental hCG (human chorionic gonadotrophin), later fetal LH (luteinising hormone))	Wolffian duct to epididymis, vas, seminal vesicle (weeks 8–12)	Appendix epididymis, paradidymis
DHT (dihydrotestosterone)	Formed locally by 5α-reductase type 2	External genitalia, prostate, scrotum (fused by 10–12 weeks)	–
INSL3 (insulin-like peptide 3)	Leydig	Gubernacular swelling (via RXFP2 receptor)	–

Physiological descent

- At **7–8 weeks** the testis lies near the kidney, held by the **CSL (cranial suspensory ligament)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks and attaches to the epididymis.

	Transabdominal phase	Inguinoscrotal phase
Timing	Weeks 10–15	From the 7th month; passage rarely before 22 weeks; most testes scrotal after 27 weeks
Events	**CSL regresses**; **gubernaculum swells**; its cord shortens, holding the testis at the internal ring while the kidney moves cranially	Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; bulb then regresses and processus closes

	Transabdominal phase	Inguinoscrotal phase
Control	CSL regression: androgen. Swelling: INSL3 (occurs even in androgen resistance)	**Androgen**, mostly via the **GFN (genitofemoral nerve)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure
Failure gives	Intra-abdominal testis	Canalicular or ectopic testis

- **Modern view:** transabdominal "descent" is largely relative. Barteczko and Jacob describe **five phases (5–28 weeks)**. The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis follow** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster myogenesis.
- Spontaneous postnatal descent occurs mainly in the first 3–6 months.

Abnormalities of descent

Type	Description
Undescended (cryptorchid)	Arrested on the normal path: abdominal, "peeping" at the ring, canalicular, prescrotal. 1–4% of term boys, up to 45% preterm; 75–80% palpable; 60–70% unilateral
Ectopic	Off the path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, contralateral scrotum
Retractile	Scrotal testis pulled up by the cremaster; stays down after sustained traction; some become acquired
Acquired (ascending)	Documented scrotal earlier; diagnosed at 8–11 years; fibrous processus remnant
Recurrent, secondary, iatrogenic	Re-ascent after spontaneous descent; after hernia repair or orchidopexy
Vanishing vs agenesis	Lost after formation (perinatal torsion; monorchia, anorchia) vs never formed

Failed step	Result
CSL retained	Intra-abdominal testis
Abnormal or ectopic gubernacular attachment	Canalicular or ectopic testis
Processus not closed	Hernia, hydrocele (patent in 87% of unilateral cases)
Processus not fully involuted	Acquired cryptorchidism

Associated anomalies and sequelae

- **Testicular: TTE (transverse testicular ectopia)**, polyorchidism, splenogonadal fusion; epididymal non-fusion.
- **PMDS (persistent Müllerian duct syndrome):** mutation of AMH or its receptor (AMHR2); bilateral cryptorchidism or TTE.
- **Genital and DSD (differences of sex development):** hypospadias (12–24%); bilateral non-palpable testes need karyotype and 17-OHP (17-hydroxyprogesterone) to exclude **CAH (congenital adrenal hyperplasia)**.
- **Syndromes:** prune belly (all), Klinefelter, androgen insensitivity, hypogonadotrophic hypogonadism, cerebral palsy, gastroschisis, omphalocele, myelomeningocele, PUV (posterior urethral valves).
- **Sequelae:** subfertility (worse if bilateral), **TGCT (testicular germ cell tumour) 2–5-fold**, torsion, hernia.

- **Treatment in brief:** orchidopexy from 6 months, by 18 months; laparoscopy for non-palpable; no imaging, no hormones.

Diagram to draw: sagittal fetus showing CSL, gubernaculum, processus vaginalis and both phases.

> **Key points to remember:**
>
> - SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
> - Transabdominal: CSL regression and INSL3 swelling; inguinoscrotal: androgen via GFN and CGRP.
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Most testes scrotal after 27 weeks.

Q2. Describe staging of CA prostate and management of nonorgan confined CA prostate

Type: LAQ (25 marks) (also answered in Practice Set 10 Paper 2, Q1)

Staging: TNM (AJCC (American Joint Committee on Cancer) 8th edition, 2017)

Stage	Definition
T1	Not palpable: T1a incidental in ≤5% of resected tissue; T1b >5%; T1c found on needle biopsy
T2	Palpable, confined: T2a ≤ half of one lobe; T2b > half of one lobe; T2c both lobes
T3	**T3a** extraprostatic extension (including microscopic bladder neck invasion); **T3b** seminal vesicle invasion
T4	Fixed or invading external sphincter, rectum, bladder, levators, pelvic wall
N	N0; **N1 regional (pelvic) nodes**
M	**M1a** non-regional nodes; **M1b** bone; **M1c** other sites

- **Traps:** seminal vesicle invasion is T3b, not T4; non-regional nodes are M1a.
- **ISUP (International Society of Urological Pathology) grade groups (GG):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.
- **Risk groups (EAU (European Association of Urology)):** high risk = PSA (prostate-specific antigen) >20 ng/mL, GG4–5 or cT2c; **locally advanced = cT3–4 or cN1**. AJCC prognostic groups combine T, N, M, PSA and grade.

Staging work-up

- DRE (digital rectal examination) sets clinical T stage but under- and over-stages. PSA predicts extent: organ-confined about 80% if PSA <4, under 50% if >10; nodes in about 75% if >50 ng/mL.
- **mpMRI (multiparametric magnetic resonance imaging):** extraprostatic extension and seminal vesicle invasion; best specificity in higher-risk men.
- **High risk:** bone scan plus CT (computed tomography) or MRI; **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**. **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Nomograms (Partin, Kattan, CAPRA (Cancer of the Prostate Risk Assessment)) and genomic classifiers (Decipher) refine risk.

Management principle

- Non-organ-confined disease (T3–4 and/or N1, M0) has a high chance of micrometastases. **Single-modality treatment usually fails**, so treatment is **multimodal**: local therapy plus systemic ADT (androgen deprivation therapy).

A. Locally advanced (cT3–4 N0 M0)

1. EBRT (external beam radiotherapy, RT) plus long-term ADT: standard

- Dose-escalated IMRT (intensity-modulated radiotherapy) with image guidance, about 76–78 Gy, with **ADT for 18–36 months** (NCCN (National Comprehensive Cancer Network); EAU 2–3 years). Neoadjuvant ADT shrinks the target and is synergistic with radiation.
- **EORTC 22863:** 3 years of ADT with RT raised 5-year overall survival (78% vs 62%). **RTOG 92-02:** 28 vs 4 months; survival gain in Gleason 8–10.
- **SPCG-7 and MRC PR07/NCIC PR.3:** adding RT to ADT reduced cancer death (RR (relative risk) 0.44; HR (hazard ratio) 0.46). ADT alone is inferior.
- **Brachytherapy boost** improves biochemical control (ASCENDE-RT).

- **Very high risk:** add **abiraterone for 2 years** (STAMPEDE M0: metastasis-free survival HR 0.53). Eligible: N1, or two of cT3–4, Gleason ≥8, PSA ≥40 ng/mL.

2. RP (radical prostatectomy) with ePLND (extended pelvic lymph node dissection): selected men

- Fit, life expectancy >10 years, resectable (mainly cT3a). Clinical overstaging occurs in 7–27%; 10-year disease-specific survival about 80–90%.
- Neoadjuvant ADT downstages clinically but gives **no survival gain**.
- **After surgery:** pT3 or positive margin: **observation with early salvage RT** (RADICALS-RT, RAVES, GETUG-AFU 17); adjuvant RT only for very high-risk pathology (GG4–5, pT3b, margins).

3. Other options

- **ADT alone:** only for men unfit or unwilling for local therapy; immediate if symptomatic, high PSA or rapid doubling.
- **Docetaxel with RT and ADT:** RTOG 0521 overall survival HR 0.69; not routine.
- Cryotherapy or HIFU (high-intensity focused ultrasound): investigational in high risk.

B. Node-positive (N1 M0)

- **cN1:** RT to prostate and pelvic nodes with long-term ADT plus 2 years of abiraterone; RP with ePLND in selected men.
- **pN1 after prostatectomy: ECOG 7887**: immediate ADT improved overall survival; ADT plus pelvic RT beats ADT alone. Observation is an option for ≤2 nodes with undetectable PSA.

C. Metastatic (M1), in brief

- ADT plus an **ARPI (androgen receptor pathway inhibitor)** (abiraterone, enzalutamide, apalutamide) or a triplet with docetaxel; prostate RT for low-volume disease (see 2025-Winter Paper 3, Q3b).

```
Non-organ-confined, M0
 ├─ Fit, cT3a, resectable → RP + ePLND → early salvage RT ± ADT
 ├─ cT3–4 or cN1 (most) → EBRT + ADT 18–36 mo
 │       very high risk / N1 → + abiraterone 2 years
 └─ Unfit for local therapy → ADT (immediate if symptomatic)
```

Local complications and follow-up

- Bladder outlet obstruction (channel TURP (transurethral resection of the prostate)), ureteric obstruction (stent or nephrostomy), haematuria (palliative RT).
- PSA every 3–6 months; bone health, metabolic and cardiovascular monitoring on ADT.

Diagram to draw: sagittal pelvis showing T3a, T3b and T4 spread.

> **Key points to remember:**
>
> - T3b = seminal vesicle; M1a = non-regional nodes.
> - EBRT + 2–3 years ADT is standard; EORTC 22863 and SPCG-7.
> - STAMPEDE: abiraterone for 2 years in very high-risk or N1 disease.
> - After RP: early salvage RT preferred to routine adjuvant RT.

Q3a. Testicular microlithiasis

Type: SAQ (10 marks)

Definition

- **TM (testicular microlithiasis):** multiple small (<3 mm) bright echogenic foci **without acoustic shadowing** scattered in the testicular parenchyma on ultrasound.
- **Classic TM:** ≥5 microliths per testis (or per image); **limited TM:** <5, of no clinical significance.

Pathology

- Laminated calcified bodies **inside seminiferous tubules**, formed on degenerated tubular cells that Sertoli cells fail to clear. Affected tubules are often obliterated.

Prevalence

Population	Prevalence
General population	5–10% (population-based 4.2%, rising with age)
Multicentre review, >37,000 males	2.9%
Boys with cryptorchidism	3.5%
Previously cryptorchid testis vs contralateral	14% vs 19% (no difference)

Associations

- Cryptorchidism, subfertility, **GCNIS (germ cell neoplasia in situ)** and TGCT (testicular germ cell tumour), Klinefelter syndrome, Down syndrome.

Cancer significance

- Tumours were found in 4.64% with TM vs 0.33% without, but cryptorchidism was not accounted for.
- **TM is not clearly an independent risk factor** for TGCT. An EAU (European Association of Urology) paediatric panel review of several hundred children found only one cancer through puberty.
- **Exception:** contralateral TM in a man with previous GCT (germ cell tumour) signals a higher GCNIS risk.

Management

- **Isolated TM, no risk factors:** reassure; **no further evaluation, no routine ultrasound follow-up**.
- **TM with risk factors** (previous GCT, cryptorchidism, infertility, atrophic testis): teach **monthly self-examination from puberty**; ultrasound only for a palpable change.
- **Biopsy:** not for TM alone. Contralateral biopsy at orchidectomy is considered when GCNIS risk is high: atrophic testis (<12 mL), cryptorchidism, age <40.
- Counsel carefully: incidental TM causes anxiety and unnecessary tests.

Differential diagnosis on ultrasound

- Coarse calcification, "burned-out" tumour scar, stellate scar (these need orchidectomy if intratesticular), scrotal pearls, rete testis ectasia.

Also know: routine screening ultrasound for infertility is discouraged because of incidental TM; follow-up schedules remain undefined.

Diagram to draw: ultrasound sketch of a testis with scattered non-shadowing echogenic dots.

> **Key points to remember:**
>
> - Classic TM ≥5 microliths; limited <5.
> - Not an independent cancer risk factor in isolation.
> - Self-examination only with risk factors; no routine imaging.

Q3b. Postoperative lymphatic leak

Type: SAQ (10 marks)

Taken as lymphocele and chylous ascites after RPLND (retroperitoneal lymph node dissection), nephrectomy, pelvic node dissection and transplantation.

Types and settings

Type	Definition	Setting and frequency
Lymphocele	Lymph collection **without epithelial lining**	**PLND (pelvic lymph node dissection):** radiological up to 60%, significant 2–5%. **Transplant:** symptomatic 0.03–26%, peak 6 weeks. **RPLND:** ≤1.7%
Chylous ascites	Lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks	RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection

- **Risk factors:** suprahilar dissection near the cisterna chyli (L1–L2), extended templates, post-chemotherapy surgery; at transplant: mTOR (mammalian target of rapamycin) inhibitors, obesity, re-transplant.

Clinical features

- **Lymphocele:** usually silent. Large ones compress the iliac vein (leg oedema, DVT (deep vein thrombosis)), bladder (frequency, pelvic pressure), ureter (hydronephrosis, graft dysfunction); may become infected.
- **Chylous ascites:** rising girth, early satiety, dyspnoea; milky drain fluid.

Diagnosis

- **Ultrasound or CT (computed tomography):** thin-walled collection; air or rim enhancement means infection.
- **Aspirate:** lymph creatinine equals serum (urinoma: much higher). Chyle: **triglyceride 2–8 × serum**, protein >3 g/dL.

Management

Lymphocele (treat by symptoms, not size alone)

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Asymptomatic, <5 cm → observe
Symptomatic → percutaneous drain (± sclerotherapy)
Recurrent / thick-walled → laparoscopic peritoneal fenestration (best)
Infected → drainage + antibiotics; exclude urine leak
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- Aspiration alone recurs; sclerosants (povidone-iodine, doxycycline) may hinder later fenestration.

Chylous ascites (stepwise)

1. Paracentesis; drain if it reaccumulates.
2. **Very low-fat, high-protein diet with MCT (medium-chain triglycerides)**, which are absorbed directly into portal blood, bypassing lacteals; plus **octreotide**.
3. Nil by mouth with **TPN (total parenteral nutrition)**.
4. Refractory: lymphangiography with embolisation, peritoneovenous shunt or surgical ligation.

- Continue a low-fat diet for **≥4 weeks** after resolution.

Prevention

- **Clip or ligate lymphatics** (cephalad at the renal hilum in RPLND; perivascular at transplant); avoid unneeded suprahilar dissection; peritoneal flap after robotic PLND.

Diagram to draw: pelvic lymphocele compressing the iliac vein and bladder.

> **Key points to remember:**
>
> - Lymphocele: commonest PLND complication; laparoscopic fenestration is best.
> - Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
> - Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Q3c. Hyperfiltration injury in AKI and CKD

Type: SAQ (10 marks)

Definition

- **Hyperfiltration:** a rise in **SNGFR (single-nephron glomerular filtration rate)** in surviving nephrons when nephron mass falls; whole-kidney GFR (glomerular filtration rate) may look normal.
- **Brenner hypothesis (1982):** the adaptation itself damages glomeruli.

Physiology

- Net filtration pressure ≈ 10 mmHg (hydrostatic 60 − Bowman's 18 − oncotic 32).
- Afferent dilatation raises flow and filtration; **efferent constriction (angiotensin II) raises filtration pressure**.

Mechanism

```
Nephron loss → afferent dilatation + efferent constriction
→ raised glomerular capillary pressure and flow (hyperfiltration)
→ glomerular hypertrophy; podocytes cannot divide → stretch, detachment
→ proteinuria → FSGS (focal segmental glomerulosclerosis)
→ tubulointerstitial inflammation, fibrosis → more nephron loss (vicious cycle)
```

In AKI (acute kidney injury)

- Incomplete repair leaves fewer nephrons, which hyperfilter; creatinine may normalise while **renal reserve is lost**.
- This drives the **AKI-to-CKD (chronic kidney disease) transition**; AKI persisting ≥90 days becomes CKD.
- Each perioperative AKI episode is a **modifiable risk factor** for CKD progression.

In CKD

- Final common pathway of progression; marked by **albuminuria (KDIGO (Kidney Disease: Improving Global Outcomes) A stages)**, hypertension, falling eGFR (estimated GFR).

Urological settings

Setting	Point
Nephrectomy or donation	Compensatory hypertrophy 20–30%; donor function 60–80% of pre-donation
Partial nephrectomy, solitary kidney	>50% parenchymal loss → proteinuria, FSGS; marked when total nephron mass falls >80%; ESRD (end-stage renal disease) if <20–30% of one kidney remains
Congenital solitary kidney, MCDK (multicystic dysplastic kidney)	Monitor BP (blood pressure), proteinuria, yearly ultrasound

- **Also know:** reflux nephropathy, infundibulopelvic stenosis, diabetes, obesity, pregnancy, low birth weight (low nephron endowment).

Prevention and treatment

- Nephron sparing; avoid NSAIDs (non-steroidal anti-inflammatory drugs) and nephrotoxins.
- **BP <130/80 mmHg**; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)** in proteinuric disease: efferent dilatation lowers

glomerular pressure.

- **SGLT2 (sodium–glucose cotransporter-2) inhibitors:** restore tubuloglomerular feedback, causing afferent constriction. **DAPA-CKD** and **EMPA-KIDNEY** slowed progression with or without diabetes.
- Finerenone (type 2 diabetic CKD); dietary protein restriction.

Diagram to draw: glomerulus with dilated afferent and constricted efferent arterioles.

> **Key points to remember:**
>
> - Hyperfiltration = raised SNGFR; Brenner 1982.
> - Leads to FSGS, proteinuria and the AKI-to-CKD transition.
> - ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

Q3d. Pelvicalyceal anomalies other than PUJ obstruction

Type: SAQ (10 marks)

Introduction

- Congenital pelvicalyceal variants occur in about 4% of people.
- **Key question:** is there obstruction? Dilatation without obstruction needs no treatment; stasis causes stones and infection.

Anomalies

Anomaly	Features	Management
Calyceal diverticulum	Transitional-lined intrarenal cavity joined to a calyx (Type I, commoner, small) or pelvis (Type II, larger, symptomatic) by a narrow neck; 4.5 per 1000; congenital (persistent 3rd–4th generation ureteric branches) or acquired. **Stones in almost 40%**; milk of calcium	Asymptomatic: observe. Pain, infection, stones: ureteroscopic neck incision, **percutaneous ablation with fulguration**, laparoscopic marsupialisation; SWL (shock wave lithotripsy) ineffective
Hydrocalycosis	Dilated major calyx connected to the pelvis; infundibular obstruction by vessel, stenosis, scarring or achalasia	Symptomatic only: infundibulotomy or dilatation
Megacalycosis	**Non-obstructive**; malformed papillae; **12–20 calyces**; pelvis normal; male 6:1; normal diuretic renogram	**None**; may coexist with segmental megaureter
Infundibular / infundibulopelvic stenosis	Narrow infundibula ± small pelvis; usually bilateral, with VUR (vesicoureteric reflux), dysplasia; over a third reach renal failure (hyperfiltration)	Monitor creatinine and eGFR (estimated glomerular filtration rate); ACE (angiotensin-converting enzyme) inhibitors; surgery only if obstruction harms function: endoscopic, calicocalicostomy, ureterocalicostomy
Extrarenal pelvis	Large pelvis outside the sinus; mimics dilatation	None
Bifid pelvis	About 10%; normal variant	None
Unipapillary kidney	Single papilla and calyx; rare; usually hypoplastic kidney	None

- **Also know:** acquired infundibular stenosis (tuberculosis, post-PCNL (percutaneous nephrolithotomy)); Fraley syndrome (crossing artery on upper infundibulum).

Evaluation

- **Ultrasound:** cyst-like space; milk of calcium layers and shifts with position.
- **CT (computed tomography) urography or MRU (magnetic resonance urography): delayed contrast pooling** in a diverticulum.
- **Diuretic renogram (MAG3 (mercaptoacetyltriglycine)):** separates obstruction (hydrocalycosis) from non-obstruction (megacalycosis, extrarenal pelvis).
- **Retrograde pyelography:** maps the neck.

Differential diagnosis

- Obstructive hydronephrosis, papillary necrosis, pyelonephritic clubbing, tuberculosis, renal cyst.

- **Stones:** retrograde first (find, dilate or laser-incise the ostium); percutaneous for lower-pole diverticula.

Diagram to draw: kidney showing diverticulum with narrow neck, hydrocalyx and megacalyces.

> **Key points to remember:**
>
> - Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
> - Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
> - Infundibulopelvic stenosis: bilateral, renal failure risk.

Q3e. TURP instruments

Type: SAQ (10 marks) (irrigation fluids also answered in 2017-Summer Paper 2, Q3a)

Introduction

- **TURP (transurethral resection of the prostate):** an electrified loop removes transition-zone adenoma from bladder neck to verumontanum, down to the surgical capsule.

Resectoscope

Part	Description
Outer sheath	24–28 Fr; **insulated beak** prevents arcing to the urethra
Inner sheath	Creates **continuous flow**: inflow through the inner, outflow between sheaths (passive or suction). Lower bladder pressure, clear vision, no repeated emptying
Obturator	Blunt or visual; atraumatic passage
Working element	Moves the loop. **Passive:** spring holds loop inside sheath; cutting stroke by spring recoil. **Active:** surgeon's thumb pulls the extended loop in
Telescope	**Hopkins rod-lens**, usually **30°** (12° alternative)

Electrodes

- **Cutting loop:** tungsten (monopolar); **platinum–iridium (bipolar plasma)**.
- Roller ball or barrel (coagulation), vaporisation button, **Collins knife** (TUIP (transurethral incision of the prostate)).

Energy

- **Generator:** about 500 kHz, too fast to stimulate nerves.
- **Cut:** continuous waveform. **Coagulation:** interrupted, higher voltage. **Blend:** better haemostasis, cuts less well.

Monopolar	Bipolar
Loop → patient → **dispersive pad**	Active and return poles on the loop
Non-conducting, hypo-osmolar irrigant: glycine 1.5%, sorbitol, water	**Normal saline**
TUR (transurethral resection) syndrome risk	Virtually none

Imaging and irrigation

- **Light source:** xenon or LED (light-emitting diode) with fibreoptic cable; camera and monitor.
- **Irrigant** at body temperature; bag about **60 cm** high (each extra 10 cm more than doubles absorption).

Accessories

- **Ellik evacuator or Toomey syringe** for chips.
- Van Buren sounds or Otis urethrotome: dilate meatus 2 Fr above sheath size (prevents stricture).
- **Three-way Foley catheter** (22–24 Fr) with catheter guide; continuous irrigation set.

Troubleshooting a loop that will not cut

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Scope–generator connection → foot pedal →
irrigant matches generator (glycine for monopolar, saline for bipolar) →
patient grounding (pad contact)
```

Recent advances

- Bipolar plasma resection; chip-on-tip digital optics; laser enucleation scopes; single-use resectoscopes.

Diagram to draw: assembled resectoscope labelling sheaths, working element, loop and flow channels.

> **Key points to remember:**
>
> - Continuous-flow 24–28 Fr sheath, 30° telescope.
> - Tungsten loop (monopolar) vs platinum–iridium (bipolar).
> - Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.

Q3f. Methods to estimate prostate size

Type: SAQ (10 marks) (TRUS also answered in 2017-Winter Paper 1, Q3e)

Why measure

- Chooses treatment: 5ARI (5α-reductase inhibitor) if >30 mL (AUA (American Urological Association)) or >40 mL (EAU (European Association of Urology)); type of BPH (benign prostatic hyperplasia) surgery.
- **PSAD (PSA density)** = PSA (prostate-specific antigen) ÷ volume; >0.15 suggests cancer. Also AUR (acute urinary retention) risk, brachytherapy planning.
- Normal adult gland 20–25 g; about 48 g at 60–70 years. **1 cm³ ≈ 1 g**.

Methods

Method	Technique	Accuracy and use
DRE (digital rectal examination)	Estimate size, consistency	**Underestimates** above 30 mL (lateral lobes impalpable); flat gland predicts <30 mL
PSA as surrogate	Age-specific cut-offs for **>40 mL:** >1.6 ng/mL (50s), >2.0 (60s), >2.3 (70s)	AUC (area under curve) 0.76–0.78; within ±20% of TRUS in most
TAUS (transabdominal ultrasound)	Full bladder; transverse width and height, sagittal length	Less precise than TRUS; also PVR (post-void residual) and **IPP (intravesical prostatic protrusion)**
TRUS (transrectal ultrasound)	Ellipsoid formula; planimetry	**Method of choice**
MRI (magnetic resonance imaging)	T2-weighted ellipsoid or segmentation (manual or automated)	Most accurate and reproducible
CT (computed tomography)	Axial images	Overestimates; poor capsule definition
Cystoscopy	Urethral length, lobes, median lobe	Rough; guides procedure choice

TRUS formulas

Shape assumed	Formula
Ellipsoid (standard)	**π/6 (0.52) × transverse × AP (anteroposterior) × length**
Sphere	π/6 × transverse³
Prolate spheroid	π/6 × transverse² × AP
Planimetry	Sum of serial transverse areas

- Transverse and AP at the widest axial section; length sagittally. All formulas correlate >0.90 with specimen weight.
- **Planimetry:** probe on a stepper, images every 3–5 mm, areas summed; most precise; used for brachytherapy.

IPP grading (suprapubic, sagittal, bladder 100–200 mL)

- Grade I <5 mm, II 5–10 mm, III >10 mm; ≥10 mm predicts BOO (bladder outlet obstruction) and failed trial without catheter.

Diagram to draw: TRUS transverse and sagittal views with three measured diameters.

Key points to remember:

- Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
- DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
- Planimetry most precise; MRI most reproducible.

Rapid Revision Sheet – Practice Set 14 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent. (LAQ)

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression and INSL3 swelling; inguinoscrotal: androgen via GFN and CGRP.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Most testes scrotal after 27 weeks.

Q2. Describe staging of CA prostate and management of nonorgan confined CA prostate (LAQ)

- T3b = seminal vesicle; M1a = non-regional nodes.
- EBRT + 2–3 years ADT is standard; EORTC 22863 and SPCG-7.
- STAMPEDE: abiraterone for 2 years in very high-risk or N1 disease.
- After RP: early salvage RT preferred to routine adjuvant RT.

Q3a. Testicular microlithiasis (SAQ)

- Classic TM ≥5 microliths; limited <5.
- Not an independent cancer risk factor in isolation.
- Self-examination only with risk factors; no routine imaging.

Q3b. Postoperative lymphatic leak (SAQ)

- Lymphocele: commonest PLND complication; laparoscopic fenestration is best.
- Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
- Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Q3c. Hyperfiltration injury in AKI and CKD (SAQ)

- Hyperfiltration = raised SNGFR; Brenner 1982.
- Leads to FSGS, proteinuria and the AKI-to-CKD transition.
- ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

Q3d. Pelvicalyceal anomalies other than PUJ obstruction (SAQ)

- Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
- Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
- Infundibulopelvic stenosis: bilateral, renal failure risk.

Q3e. TURP instruments (SAQ)

- Continuous-flow 24–28 Fr sheath, 30° telescope.
- Tungsten loop (monopolar) vs platinum–iridium (bipolar).
- Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.

Q3f. Methods to estimate prostate size (SAQ)

- Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
- DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
- Planimetry most precise; MRI most reproducible.